

# Saran D

## Machine Learning Engineer

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Chennai, Tamil Nadu



Machine Learning Engineer fascinated by developing models for computer vision, natural language processing, and time series forecasting. Skilled in data collection, analysis, SQL querying, and modeling, with a track record of deploying secure APIs and applications on AWS and Linux servers. Adept at using various machine learning techniques to solve complex problems and deliver actionable insights

### Experience

Tamil Nadu e-Governance Agency

12/2023 - present

Designation: Machine Learning Engineer -CV(Jr.)

- **Oral Precancer Detection:** Developed a custom YOLO model using the dentist's bounding box annotated dataset to identify oral precancer. performed in hyperparameter tuning to improve accuracy.
- **Transliteration Model:** Implemented a bi-directional transliteration model between Tamil and English using Fairseq, incorporating fuzzy algorithms for name and sound matching.
- **Face Recognition and Anti-Spoofing:** Built an end-to-end face recognition model along with a separate anti-spoofing detection system, capturing facial landmarks for enhanced accuracy.
- **Chatbot Development:** Designed a rule-based chatbot proof-of-concept using the RASA framework and also developed a conversational chatbot tailored to a domain-specific dataset by using langchain.
- **Image Annotation Automation:** Integrated the SAM model into an open-source annotation tool to automate polygon annotations and converting to binary masks, reducing manual annotation effort for image datasets.
- **Crop Detection:** Developed a crop detection model for banana and coconut using YOLO segmentation, achieving over **70% accuracy** on real-world datasets.
- **Pest Identification:** Built a model for identifying pests such as **leaf folder, fall armyworm, and stem borer** in plants, providing remedial recommendations based on expert-annotated real-world field images.
- **Electricity Demand Forecasting:** Developed an electricity demand forecasting model for TANGEDCO, utilizing 15-minute interval data and correlated parameters such as temperature, humidity, and special events. Applied statistical and deep learning techniques, achieving a Mean Absolute Percentage Error (MAPE) of around **8%**.

**Key Contributions:**

- Led data preprocessing, hyperparameter tuning, and model deployment efforts for multiple projects.
- Conducted extensive data collection and analyzed various energy sources to study their impact on power demand in Tamil Nadu, identifying trends and seasonality patterns.
- Implemented forecasting techniques such as moving averages, trailing windows, and feature engineering to predict electricity demand using models like LSTM and Prophet.
- Successfully migrated databases using PySpark, resulting in a 40% reduction in processing time and resource utilization.
- Developed a beta version of a web application using Node.js, Express.js, and MySQL, supporting agricultural stakeholders through seamless data input and retrieval.
- Built and maintained secure APIs for data sharing, including IP whitelisting and a dashboard for monitoring API logs.
- Developed and deployed scalable applications using AWS EC2 instances and Elastic Beanstalk, supporting projects like the Women's Entitlement Project.

**Data Science and Business Analytics - Internship**

*07/2021 - 08/2021*

- Build a Linear Regression model to Predict the Student's Mark

**Frontend Developer - Internship**

*12/2020 - 01/2021*

- Implemented a simple dynamic jQuery scroller that smoothly changed the background color while scrolling, enhancing the website's interactive features
- Developed a project using HTML, CSS, JavaScript, and Bootstrap.

**Education**

- **Master of Computer Application [MCA]**  
S.A.Engineering College, Thiruverkadu

*2020 - 2022*

**Skills:**

- **Programming Languages:** Python, JavaScript
- **Web Development:** Node.js, EJS
- **Cloud Platforms:** AWS (EC2, Elastic Beanstalk), TNSDC
- **Database Management:** PostgreSQL, MySQL, MongoDB
- **Data Processing & Analysis:** PySpark, Pandas, NumPy
- **Machine Learning & AI:** OpenCV, Fairseq, YOLO, LSTM, RASA
- **Version Control & Deployment:** Git, PM2, Docker